



Carry out vehicle maintenance or minor repairs

D2.TTO.CL4.02

Assessor Manual



Carry out vehicle maintenance or minor repairs

D2.TTO.CL4.02

Assessor Manual



William
Angliss
Institute

Specialist centre
for foods, tourism
& hospitality

Project Base

William Angliss Institute of TAFE
555 La Trobe Street
Melbourne 3000 Victoria
Telephone: (03) 9606 2111
Facsimile: (03) 9670 1330

Acknowledgements

Project Director:	Wayne Crosbie
Project Manager	Jim Irwin
Chief Writer:	Alan Hickman
Subject Writer:	Alan Hickman
Editor:	Jim Irwin
DTP/Production:	Daniel Chee, Mai Vu, Cindy Curran

The Association of Southeast Asian Nations (ASEAN) was established on 8 August 1967. The Member States of the Association are Brunei Darussalam, Cambodia, Indonesia, Lao PDR, Malaysia, Myanmar, Philippines, Singapore, Thailand and Viet Nam.

The ASEAN Secretariat is based in Jakarta, Indonesia.

General Information on ASEAN appears online at the ASEAN Website: www.asean.org.

All text is produced by William Angliss Institute of TAFE for the ASEAN Project on "Toolbox Development for Tourism Labour Divisions for Travel Agencies and Tour Operations".

This publication is supported by the Australian Government's aid program through the ASEAN-Australia Development Cooperation Program Phase II (AADCP II).

Copyright: Association of Southeast Asian Nations (ASEAN) 2015.

All rights reserved.

Disclaimer

Every effort has been made to ensure that this publication is free from errors or omissions. However, you should conduct your own enquiries and seek professional advice before relying on any fact, statement or matter contained in this book. The ASEAN Secretariat and William Angliss Institute of TAFE are not responsible for any injury, loss or damage as a result of material included or omitted from this course. Information in this module is current at the time of publication. Time of publication is indicated in the date stamp at the bottom of each page.

Some images appearing in this resource have been purchased from stock photography suppliers Shutterstock and iStockphoto and other third party copyright owners and as such are non-transferable and non-exclusive. Clip arts, font images and illustrations used are from the Microsoft Office Clip Art and Media Library. Some images have been provided by and are the property of William Angliss Institute.

Additional images have been sourced from Flickr and SXC and are used under Creative Commons licence: <http://creativecommons.org/licenses/by/2.0/deed.en>

File name: AM_Carry_out_vehicle_maintenance_minor_repairs_270715

Table of contents

Competency Based Assessment (CBA) – An Introduction for Assessors.....	1
Competency standard.....	11
Oral questions.....	23
Written questions	33
Answers to written questions	49
Observation checklist.....	67
Third Party Statement.....	71
Competency recording sheet	73

Competency Based Assessment (CBA) – An Introduction for Assessors

Assessment is the process of identifying a participant's current knowledge, skills and attitudes sets against all elements of competency within a unit of competency.

Suggested Assessment Methods

For each unit of competency a number of assessment tools have been identified including:

- Work Projects
- Oral Questions
- Written Questions
- Third Party Statements
- Observation Checklists.

Instructions and Evidence Recording Sheets have been identified in this Assessment Manual for use by Assessors.

Alternative Assessment Methods

Whilst the above mentioned assessment methods are suggested assessment methods, the assessor may use an alternate method of assessment taking into account:

- a) The nature of the unit
- b) The strengths of participants
- c) The number of participants in the class
- d) Time required to complete assessments
- e) Time dedicated to assessment
- f) Equipment and resources required.

Alternate assessment methods include:

- Practical demonstrations
- Practical demonstrations in simulated work conditions
- Problem solving
- Portfolios of evidence
- Critical incident reports
- Journals
- Oral presentations
- Interviews
- Videos
- Visuals/slides/audiotapes
- Case studies

- Log books
- Projects and Role plays
- Group projects
- Recognition of Prior Learning.

Whilst there is no specific instruction or evidence collection documents for all the alternative assessment methods, assessors can record competency in the 'Other' section within the 'Competency Recording Sheet'.

Selection of Assessment Methods

Each assessor will determine the combination of Assessment Methods to be used to determine Competency for each Competency Unit on a student by student basis.

'Sufficient' evidence to support the 'Pass Competent'/'Not Yet Competent' decision must be captured.

In practice this means a minimum of 2 – 3 Assessment Methods for each candidate for each Competency Element is suggested.

At least one method should provide evidence of practical demonstration of competence.

The following assessment methods deemed to provide evidence of practical demonstration of competence include:

- Practical Work Projects
- Third Party Statement
- Observation Checklist.

Assessing Competency

Competency based assessment does not award grades, but simply identifies if the participant has the knowledge, skills and attitudes to undertake the required task to the specified standard.

Therefore, when assessing competency, an assessor has two possible results that can be awarded:

- 'Pass Competent' (PC)
- 'Not Yet Competent' (NYC).

Pass Competent (PC)

If the participant is able to successfully answer or demonstrate what is required, to the expected standards of the performance criteria, they will be deemed as 'Pass Competent' (PC).

The assessor will award a 'Pass Competent' (PC) if they feel the participant has the necessary knowledge, skills and attitudes in all assessment tasks for a unit.

Not Yet Competent' (NYC)

If the participant is unable to answer or demonstrate competency to the desired standard, they will be deemed to be 'Not Yet Competent' (NYC).

This does not mean the participant will need to complete all the assessment tasks again. The focus will be on the specific assessment tasks that were not performed to the expected standards.

The participant may be required to:

- a) Undertake further training or instruction
- b) Undertake the assessment task again until they are deemed to be 'Pass Competent'.

Regional Qualifications Framework and Skills Recognition System

The 'Regional Qualifications Framework and Skills Recognition System', also known as the 'RQFSRS' is the overriding educational framework for the ASEAN region.

The purpose of this framework is to provide:

- A standardised teaching and assessment framework
- Mutual recognition of participant achievement across the ASEAN region. This includes achievement in individual Units of Competency or qualifications as a whole.

The role of the 'RQFSRS' is to provide, ensure and maintain 'quality assurance' across all countries and educational providers across the ASEAN region.

Recognition of Prior Learning (RPL)

Recognition of Prior Learning is the process that gives current industry professionals who do not have a formal qualification, the opportunity to benchmark their extensive skills and experience against the standards set out in each unit of competency/subject.

This process is a learning and assessment pathway which encompasses:

- Recognition of Current Competencies (RCC)
- Skills auditing
- Gap analysis and training
- Credit transfer.

Code of Practice for Assessors

This Code of Practice provides:

- Assessors with direction on the standard of practice expected of them
- Candidates with assurance of the standards of practice expected of assessors
- Employers with assurance of the standards maintained in the conduct of assessment.

The Code detailed below is based on the International Code of Ethics and Practice (The National Council for Measurement in Education [NCME]):

- The differing needs and requirements of the person being assessed, the local enterprise and/or industry are identified and handled with sensitivity
- Potential forms of conflict of interest in the assessment process and/or outcomes are identified and appropriate referrals are made, if necessary
- All forms of harassment are avoided throughout the planning, conducting, reviewing and reporting of the assessment outcomes
- The rights of the candidate are protected during and after the assessment

- Personal and interpersonal factors that are not relevant to the assessment of competency must not influence the assessment outcomes
- The candidate is made aware of rights and process of appeal
- Evidence that is gathered during the assessment is verified for validity, reliability, authenticity, sufficiency and currency
- Assessment decisions are based on available evidence that can be produced and verified by another assessor
- Assessments are conducted within the boundaries of the assessment system policies and procedures
- Formal agreement is obtained from both the candidate and the assessor that the assessment was carried out in accordance with agreed procedures
- The candidate is informed of all assessment reporting processes prior to the assessment
- The candidate is informed of all known potential consequences of decisions arising from an assessment, prior to the assessment
- Confidentiality is maintained regarding assessment results
- The assessment results are used consistently with the purposes explained to the candidate
- Opportunities are created for technical assistance in planning, conducting and reviewing assessment procedures and outcomes.

Instructions and Checklist for Assessors

Instructions

General instructions for the assessment:

- Assessment should be conducted at a scheduled time that has been notified to the candidate
- Facilitators must ensure participants are made aware of the need to complete assessments and attend assessment sessions
- If a participant is unable to attend a scheduled session, they must make arrangements with the Assessor to undertake the assessment at an alternative time
- At the end of the assessment the Assessor must give feedback and advise the participant on their PC/NYC status
- Complete the relevant documentation and submit to the appropriate department.

Preparation

- Gain familiarity with the Unit of Competency, Elements of Competency and the Performance Criteria expected
- Study details assessment documentation and requirements
- Brief candidate regarding all assessment criteria and requirements.

Briefing Checklist

- Begin the assessment by implementing the following checklist and then invite the candidate to proceed with assessment.

Checklist for Assessors

Prior to the assessment I have:	Tick (✓)	Remarks
Ensured the candidate is informed about the venue and schedule of assessment.		
Received current copies of the performance criteria to be assessed, assessment plan, evidence gathering plan, assessment checklist, appeal form and the company's standard operating procedures (SOP).		
Reviewed the performance criteria and evidence plan to ensure I clearly understood the instructions and the requirements of the assessment process.		
Identified and accommodated any special needs of the candidate.		
Checked the set-up and resources for the assessment.		
During the assessment I have:		
Introduced myself and confirmed identities of candidates.		
Put candidates at ease by being friendly and helpful.		
Explained to candidates the purpose, context and benefits of the assessment.		
Ensured candidates understood the assessment process and all attendant procedures.		
Provided candidates with an overview of performance criteria to be assessed.		
Explained the results reporting procedure.		
Encouraged candidates to seek clarifications if in doubt.		
Asked candidates for feedback on the assessment.		
Explained legal, safety and ethical issues, if applicable.		
After the assessment I have:		
Ensured candidate is given constructive feedback.		
Completed and signed the assessment record.		
Thanked candidate for participating in the assessment.		

Instructions for Recording Competency

Specifications for Recording Competency

The following specifications apply to the preparation of Evidence Gathering Plans:

- A Competency Recording Sheet must be prepared for each candidate to ensure and demonstrate all Performance Criteria and Competency Elements are appropriately assessed. This Sheet indicates how the Assessor will gather evidence during their assessment of each candidate
- This Competency Recording Sheet is located at the end of the Assessment Plan
- It is the overriding document to record competency
- The Assessor may vary the Competency Recording Sheet to accommodate practical and individual candidate and/or workplace needs
- Assessor must place a tick (✓) in the 'Assessment Method' columns to identify the methods of assessment to be used for each candidate
- Multiple Competency Elements/Performance Criteria may be assessed at the one time, where appropriate
- The assessor and participant should sign and date the Competency Recording Sheet, when all forms of evidence and assessment have been completed
- The assessor may provide and feedback or clarify questions which the participant may have in regards to the assessment grade or findings
- All documents used to capture evidence must be retained, and attached to the Competency Recording Sheet for each candidate for each Competency Unit.

Instructions for Different Assessment Methods

Specifications for Work Project Assessment

These guidelines concern the use of work projects.

The work projects identified in the Training Manuals involve a range of tasks, to be performed at the discretion of the Assessor.

Work project tasks can be completed through any form of assessment as identified in the Trainer and Trainee Manuals and stated at the start of this section.

Assessors should follow these guidelines:

- Review the Work Projects at the end of each 'Element of Competency' in the Trainee Manual to ensure you understand the content and what is expected
- Prepare sufficient resources for the completion of work activities including:
 - Time – whether in scheduled delivery hours or suggested time participants to spend outside of class hours
 - Resources – this may involve technical equipment, computer, internet access, stationery and other supplementary materials and documents
- Prepare assessment location (if done in class) making it conducive to assessment
- Explain Work Projects assessment to candidate, at the start of each Element of Competency. This ensures that participants are aware of what is expected and can collate information as delivery takes place

- Assessors can use the following phrase as a guide (where an 'X' is identified, please input appropriate information):

"At the end of each Element of Competency there are Work Projects which must be completed. These projects require different tasks that must be completed.

These work projects are part of the formal assessment for the unit of competency titled X:

 - You are required to complete these activities:
 - Using the 'X' method of assessment*
 - At 'X' location*
 - You will have 'X time period' for this assessment*
 - You are required to compile information in a format that you feel is appropriate to the assessment
 - Do you have any questions about this assessment?"
- Commence Work Project assessment:
 - The assessor may give time for participants to review the questions at this time to ensure they understand the nature of the questions. The assessor may need to clarify questions
 - Participants complete work projects in the most appropriate format
 - Participants must submit Work Project evidence to the assessor before the scheduled due date
- Assessor must assess the participant's evidence against the competency standards specified in each Element of Competency and their own understanding. The assessor can determine if the participant has provided evidence to a 'competent' standard
- Transcribe results/details to Competency Recording Sheet
- Forward/file assessment record.

Specifications for Oral Question Assessment

These guidelines concern the use of oral questioning.

Assessors should follow these guidelines.

- Prepare Assessment Record for Oral Questioning. One record for each candidate:
 - Enter Student name
 - Enter Assessor name
 - Enter Location
- Familiarise self with Questions to be asked
- Prepare assessment location (table and chairs) making it conducive to assessment
- Explain Oral Questioning assessment to candidate, using the following phrase as a guide (where a 'X' is identified, please input appropriate information):

"These oral questions are part of the formal assessment for the unit of competency titled X.

There are X questions and you are required to answer all of them to the best of your ability and I will record whether or not you have answered correctly.

We have 60 minutes for this assessment.

- I will give you feedback at the end of the assessment
- Do you have any questions about this assessment?"
- Commence Oral Questioning assessment:
 - Complete Assessment Record for the Oral Questioning by:
- a) Ticking PC or NYC, as appropriate
- b) Entering 'Remarks' as required
- c) Completing Oral Questioning within 60 minutes
- Complete Oral Questioning and provide feedback to candidate
- Transcribe results/details to Competency Recording Sheet
- Forward/file assessment record.

Specifications for Written Question Assessment

These guidelines concern the use of written questioning.

Assessors should follow these guidelines:

- Familiarise self with Questions and Answers provided
- Print and distribute copies of 'Written Questions' for participants. Ideally this should take place with adequate time for participants to answer all questions before the expected due date
- Explain Written Questioning assessment to candidate, using the following phrase as a guide (where a 'X' is identified, please input appropriate information):

"These written questions are part of the formal assessment for the unit of competency titled X.

There are X questions and you are required to answer all of them to the best of your ability.

You may refer to your subject materials, however where possible try to utilise your existing knowledge when answering questions.

Where you are unsure of questions, please ask the Assessor for further instruction. This may be answering the question orally or asking the assessor to redefine the question.

We have X time for this assessment:

- The due date for completion of this assessment is X
- On this date you must forward the completed questions to the assessor by X time on the date of X
- Do you have any questions about this assessment?"
- The assessor may give time for participants to review the questions at this time to ensure they understand the nature of the questions. The assessor may need to clarify questions
- Participants may record written answers (where possible)
- Participants must submit the written answers to the assessor before the scheduled due date

- Assessor must assess the participant's written answers against the model answers provided as a guide, or their own understanding. The assessor can determine if the participant has answered the questions to a 'competent' standard
- Transcribe results/details to Competency Recording Sheet
- Forward/file assessment record.

Specifications for Observation Checklist

These specifications apply to the use of the Observation Checklist in determining competency for candidates.

Only an approved assessor is authorised to complete the Observation Checklist.

The assessor is required to observe the participant, ideally in a simulated environment or their practical workplace setting and record their performance (or otherwise) of the competencies listed on the Observation Checklist for the Competency Unit.

To complete the Observation Checklist the Assessor must:

- Insert name of candidate
- Insert assessor name
- Insert identify of location where observations are being undertaken
- Insert date/s of observations – may be single date or multiple dates
- Place a tick in either the 'Yes' or 'No' box for each listed Performance Criteria to indicate the candidate has demonstrated/not demonstrated that skill
- Provide written (and verbal) feedback to candidate – as/if appropriate
- Sign and date the form
- Present form to candidate for them to sign and date
- Transcribe results/details to Competency Recording Sheet for candidate
- Forward/file Observation Checklist.

This source of evidence combines with other forms of assessment to assist in determining the 'Pass Competent' or 'Not Yet Competent' decision for the participant.

Specifications for Third Party Statement

These specifications relate to the use of a relevant workplace person to assist in determining competency for candidates.

The Third Party Statement is to be supplied by the assessor to a person in the workplace who supervises and/or works closely with the participant.

This may be their Supervisor, the venue manager, the Department Manager or similar.

The Third Party Statement asks the Supervisor to record what they believe to be the competencies of the participant based on their workplace experience of the participant. This experience may be gained through observation of their workplace performance, feedback from others, inspection of candidate's work etc.

A meeting must take place between the Assessor and the Third Party to explain and demonstrate the use of the Third Party Statement.

To complete the Third Party Verification Statement the Assessor must:

- Insert candidate name
- Insert name and contact details of the Third Party
- Tick the box to indicate the relationship of the Third Party to the candidate
- Present the partially completed form to the Third Party for them to finalise
- Collect the completed form from the Third Party
- Transcribe results/details to Competency Recording Sheet for candidate
- Forward/file Third Party Statement.

The Third Party must:

- Record their belief regarding candidate ability/competency as either:
 - Pass Competent = Yes
 - Not Yet Competent = No
 - Unsure about whether candidate is competent or not = Not Sure
- Meet briefly with the assessor to discuss and/or clarify the form.

This source of evidence combines with other forms of assessment to assist in determining the 'Pass Competent' or 'Not Yet Competent' decision for the candidate.

A separate Third Party Statement is required for each Competency Unit undertaken by the candidate.

Competency standard

UNIT TITLE: CARRY OUT VEHICLE MAINTENANCE OR MINOR REPAIRS		NOMINAL HOURS: 80
UNIT NUMBER: D2.TTO.CL4.02		
UNIT DESCRIPTOR: This unit deals with skills and knowledge required to provide the scheduled service and basic repairs to vehicles.		
ELEMENTS AND PERFORMANCE CRITERIA		UNIT VARIABLE AND ASSESSMENT GUIDE
Element 1: Provide scheduled service to vehicles 1.1 <i>Identify necessary minor servicing requirements for specific vehicles</i> 1.2 <i>Determine situations in which maintenance may need to be carried out</i> 1.3 <i>Undertake visual inspections of the vehicle</i> 1.4 <i>Check and adjust vehicle structure</i> 1.5 <i>Check and adjust lighting</i> 1.6 <i>Check and adjust vision-related items</i> 1.7 <i>Check and adjust entrances and exits</i> 1.8 <i>Check and adjust vehicle interior</i> 1.9 <i>Check and adjust brakes</i> 1.10 <i>Check and adjust steering</i> 1.11 <i>Check and adjust exhaust</i> 1.12 <i>Check and adjust towing connections</i>		Unit Variables The Unit Variables provide advice to interpret the scope and context of this unit of competence, allowing for differences between enterprises and workplaces. It relates to the unit as a whole and facilitates holistic assessment. This unit applies to organisations that are involved in producing brochures and marketing materials within the labour divisions of the hotel and travel industries and may include: 1. Travel Agencies 2. Tour Operation <i>Identify necessary minor servicing requirements</i> may include: • Researching manufacturer requirements in relation to routine servicing periods and items to be checked and serviced • Reading owner manual • Identifying internal vehicle service periods and requirements that may override stated manufacturer's requirements

1.13 <i>Check and adjust miscellaneous items</i> 1.14 <i>Check and adjust load restraints</i> 1.15 Check and replenish on-board emergency equipment and supplies, where necessary 1.16 <i>Make arrangements for external service provision</i> where requirements cannot be accommodated internally 1.17 Comply with <i>mandated safety requirements</i> while undertaking service Element 2: Diagnose minor vehicle faults 2.1 <i>Identify minor faults</i> 2.2 <i>Determine cause of faults</i> Element 3: Undertake minor repairs to vehicles 3.1 Remove, repair and/or refit <i>vehicle components</i> 3.2 Use correct tools and follow manufacturer's instructions 3.3 <i>Make arrangements for external repair provision</i> where requirements cannot be accommodated internally 3.4 Comply with mandated safety requirements while undertaking repairs Element 4: Complete documentation 4.1 Ensure vehicle inspection checklists are completed	• Referring to documentation relating to previous servicing that has taken place • Talking with vehicle operators to identify items requiring attention • Identifying host country legislation that applies to the maintenance of commercial vehicles used for passenger transport. <i>Vehicles</i> will include any commercial vehicle, petrol or diesel, used by the employer to support the delivery of tours and/or transfers and may include: • Cars • Four-wheel drives • Utilities • Light and heavy commercial vehicles • Combination vehicles. <i>Situations in which maintenance may need to be carried out</i> may include: • Operations conducted at day or night • Typical weather conditions • In tight or confined spaces, exposed conditions and controlled or open environments • While in the depot, base or warehouse • While in the vehicle on the road/on tour • While at the client workplace. <i>Undertake visual inspections</i> may relate to: • Looking for signs of damage • Checking for missing parts • Identifying leaking fluids • Identifying projections from the vehicle
---	---

4.2 <i>Complete internal documents to record provision of service and repairs</i> 4.3 <i>Adhere to requirements in relation to the completion and maintenance of vehicle service and repair documentation</i>	• Noting defects and matters requiring attention • Taking immediate action to rectify problems • Checking condition of operator pedals, switches, knobs and other items required to operate or control the vehicle. <i>Check and adjust vehicle structure</i> may relate to: • Panels and chassis • Vehicle stability and suspension • Roof racks, pack racks and other similar fittings. <i>Check and adjust lighting</i> may relate to: • Headlights, front and rear fog lights and driving lights • Turning indicators • Side marker lights • Brake lights • Reversing lights, including reversing warning device • Spotlights • Rear reflectors • Internal lights, including roof and individual passenger lights, map-reading lights, boot and cargo compartment lights • Dashboard and instrument illumination • Warning lights.
---	---

	<i>Check and adjust vision-related items</i> may relate to: • Glazing • Sun visors • Windscreen wipers and washing system • Internal and external rear-view mirrors • Condition of windscreen • Demisters. <i>Check and adjust entrances and exits</i> may relate to: • Doors, including hinged, sliding and pneumatic • Bonnet, boot, cargo holds and side/drop doors • Entry steps, ramps and hoists • Pneumatic supports used to keep doors open • Emergency exits. <i>Check and adjust vehicle interior</i> may relate to: • Seats and seat anchorage • Head restraints • Aisles • Seatbelts, restraints and anchor points • Airbags • Heating and ventilation • Fire protection • On-board signs and instructions
--	---

- Emergency passenger signals
 - Audible warning devices.
- Check and adjust brakes* may relate to:
- Travel of brake pedal
 - Condition of braking system and components, including fluid level
 - Parking brake and on-board secondary braking systems.
- Check and adjust steering* may relate to:
- Steering wheel, steering column, linkages and joints
 - Power steering systems, including fluid levels
 - Steering box or rack
 - Steering arms and kingpins
 - Suspension
 - Mounting points.
- Check and adjust exhaust* may relate to:
- Internal fumes
 - External smoke
 - Noise.
- Check and adjust towing connections* may relate to:
- Tow bar and chains
 - Fixed/jockey wheel
 - Draw beam
 - Electrical connections.

	<i>Check and adjust miscellaneous items</i> may relate to: • Engine, including topping up and changing engine oil • Transmission, including automatic transmission fluid level and gearbox oil level • Fuel system, including dual fuel systems and changing of filters • Electrical wiring • Water, including levels in radiator, secondary supply reservoirs, windscreen washers, flushing cooling system and adding cooling system additives, where required • Tyres, including condition, pressures, on-road tyres, spare tyres and checking on-board compressors and pumps and other items used to deflate and inflate tyres • On-board personal service facilities, including toilets, sinks, radio, television, tape systems and other items specific to individual types of vehicles • On-board communication systems, including public address systems, global positioning system, two-way radios and Emergency Position Indicating Radio Beacon (EPIRB) • Equipment for driving vehicle in specific conditions, including wheel chains, studs and rails. <i>Check and adjust load restraints</i> may relate to: • Load anchorage • Retention devices • Cab guards, headboards, sideboards and tailboards • Curtain systems • Baggage and cargo restraints. <i>Make arrangements for external service provision</i> may include: • Contacting the authorised or preferred service provider • Advising service provider of the nature of the fault or repairs required
--	--

	• Making a booking for vehicle to be serviced • Ensuring vehicle is not used until the required service has been completed, where safety is a concern. <i>Mandated safety requirements</i> will include: • Compliance with legislated requirements in relation to occupational health, safety and welfare • Compliance with external inspection requirements by authorised officers and inspectors. <i>Identify minor faults</i> may include: • Talking with vehicle operators to identify type and nature of fault, including symptoms and indicators, frequency and conditions under which fault occurs/presents • Test driving vehicle to induce fault • Running stationary vehicle to induce fault. <i>Determine cause of faults</i> may include: • Referring to service and operator manuals • Seeking advice from manufacturer, including on-line advice, over the phone assistance and in-person help • Eliminating factors from the scenario that are causing the fault on a sequential and structured basis while continuing ongoing testing • Referring to personal experience • Loading vehicle onto jacks or hydraulic lift • Using applicable sensory appraisal, including sight, feel, smell and hearing • Monitoring on-board gauges and instruments • Monitoring externally attached diagnostic equipment.
--	--

	<i>Vehicle components</i> may include: • Globes, reflectors and lenses • Fan belt and other belts • Fuses • Spark plugs and glow plugs • Mirrors • Tyres and tubes • Coolant hoses. <i>Make arrangements for external repair provision</i> may include: • Contacting the authorised or preferred service provider • Advising service provider of the nature of the repairs required • Making a booking for vehicle to be repaired • Ensuring vehicle is not used until the required repairs have been completed, where safety is a concern. <i>Complete internal documents</i> may include: • Describing service and repairs provided • Updating future scheduled service times/dates • Noting matters that will/may require attention at next service • Costing items and labour involved in service and/or repair provision. <i>Requirements in relation to the completion and maintenance of vehicle service and repair documentation</i> may include: • Host enterprise requirements • Legislated requirements
--	--

- Requirements imposed on the business by clients such as tour operators, venues and site operators who contract the use of vehicles
- Requirements imposed on the business by third parties such as insurers and auditors.

Assessment Guide

The following skills and knowledge must be assessed as part of this unit:

- Enterprise policies and procedures in regard to vehicle maintenance and minor repairs, authorisation to engage the services of external service providers, staff and customer safety and record keeping
- Knowledge of host country legislated and other requirements in relation to the service and maintenance of commercial passenger vehicles
- Principles of diagnosing vehicle faults
- Ability to use basic repair and maintenance tools, equipment and techniques
- Ability to apply knowledge of standard and basic maintenance and repair procedures
- Knowledge of vehicle manufacturer's instructions, specifications and recommended procedures for maintenance, troubleshooting and repairs
- Knowledge of safety requirements that apply when servicing and/or repairing vehicles.

Linkages To Other Units

- Implement occupational health and safety procedures
- Drive various types of service vehicles
- Establish and maintain safe touring conditions
- Drive large tour buses or coaches
- Manage operational risk
- Maintain tourism vehicles in safe and clean operational condition

	• Operate and maintain a 4-WD vehicle in safe working condition • Manage legal requirements for business compliance. Critical Aspects of Assessment Evidence of the following is essential: • Understanding of host enterprise policies and procedures in regard to vehicle maintenance and minor repairs, authorisation to engage the services of external service providers, staff and customer safety and record keeping • Understanding of host country legislated obligations that apply to the maintenance, inspection and repair of commercial passenger vehicles • Demonstrated ability to inspect and provide scheduled/routine service to a nominated vehicle type at a designated scheduled service interval • Demonstrated ability to provide effective and compliant service to a nominated vehicle type following its operation under designated adverse conditions such as sand, mud, water or off-road • Demonstrated ability to diagnose and provide effective and compliant repairs to at least three simulated, minor and unidentified faults on a nominated vehicle. Context of Assessment Assessment must ensure: • Actual or simulated workplace application of vehicle maintenance and repair skills. Resource Implications Training and assessment must include the use of real vehicles, real tools and resources and a range of vehicles requiring a range of maintenance and repair provisions; and access to workplace standards, procedures, policies, guidelines, tools and equipment. Note: vehicle faults and problems may be simulated.
--	--

Assessment Methods

The following methods may be used to assess competency for this unit:

- Observation of practical candidate performance
- Test runs of vehicles that have been serviced and/or repaired
- Portfolio of documents providing evidence of vehicle inspections, servicing and repairs
- Oral and written questions
- Third party reports completed by a supervisor
- Project and assignment work.

Key Competencies in this Unit

Level 1 = competence to undertake tasks effectively

Level 2 = competence to manage tasks

Level 3 = competence to use concepts for evaluating

Key Competencies	Level	Examples
Collecting, organising and analysing information	2	Check vehicles for faults and damage
Communicating ideas and information	1	Notify specialists regarding service and repair requirements
Planning and organising activities	1	Schedule vehicle servicing
Working with others and in teams	1	Liaise with vehicle operators to assist in fault diagnosis and identification of service and repair needs

	Using mathematical ideas and techniques	1	Calculate quantities/volumes of fluids, air and other resources needed to service and repair vehicles
	Solving problems	2	Identify and diagnose faults
	Using technology	2	Use diagnostic equipment to determine faults

Oral questions

Student name	
Assessor name	
Location/venue	
Unit of competency	Carry out vehicle maintenance or minor repairs D2.TTO.CL4.02
Instructions	1. Ask student questions from the attached list to confirm knowledge, as necessary 2. Place tick in boxes to reflect student achievement (Pass Competent 'PC' or Not Yet Competent 'NYC') 3. Write short-form student answer in the space provided for each question.

Questions	Response	
	PC	NYC
1. How do you identify the minor servicing requirements for specific vehicles?	☐	☐
2. Tell me how you determine situations in which maintenance may need to be carried out to a vehicle.	☐	☐

Questions	Response	
	PC	NYC
3. Describe what you do when conducting a visual inspection of a vehicle as part of the process of providing scheduled service.	☐	☐
4. Using an example, explain what is involved in checking and adjusting vehicle structure as part of the process of providing scheduled service.	☐	☐
5. Identify a vehicle and explain what is involved in checking and adjusting lighting for this vehicle.	☐	☐

Questions	Response	
	PC	NYC
6. Identify a vehicle and explain what is involved in checking and adjusting vision-related items.	☐	☐
7. Identify a vehicle and explain what is involved in checking and adjusting entrances and exits.	☐	☐
8. Identify a vehicle and explain what is involved in checking and adjusting vehicle interior.	☐	☐

Questions	Response	
	PC	NYC
9. Identify a vehicle and explain what is involved in checking and adjusting brakes.	☐	☐
10. Identify a vehicle and explain what is involved in checking and adjusting steering.	☐	☐
11. Identify a vehicle and explain what is involved in checking and adjusting exhaust.	☐	☐

Questions	Response	
	PC	NYC
12. Identify a vehicle and explain what is involved in checking and adjusting towing connections.	☐	☐
13. Identify a vehicle and explain what is involved in checking and adjusting miscellaneous items.	☐	☐
14. Identify a vehicle and explain what is involved in checking and adjusting load restraints.	☐	☐

Questions	Response	
	PC	NYC
15. Identify a vehicle and explain what is involved in checking and replenishing on board emergency equipment and supplies	☐	☐
16. What is involved in making arrangements for external service provision where you cannot perform the work required?	☐	☐
17. What mandated safety requirements must you comply with while undertaking service to a vehicle?	☐	☐

Questions	Response	
	PC	NYC
18. Give me three examples minor vehicle faults you have identified explaining how you became aware of these faults.	☐	☐
19. For the three examples in your answer to the previous question what did you do to diagnose each fault?	☐	☐
20. Give me an example of situations where you have removed, repaired and refitted vehicle components.	☐	☐

Questions	Response	
	PC	NYC
21. Give an example of the importance of using the correct tools and following manufacturer's instructions when maintaining tour vehicles.	☐	☐
22. Describe a situation where you have had to arrange external service for a tour vehicle which requires servicing/maintenance.	☐	☐
23. What can be the consequences of not following required safety procedures/protocols when maintaining tour vehicles?	☐	☐

Questions	Response	
	PC	NYC
24. Describe how you use a vehicle inspection checklist when maintaining tour vehicles.	☐	☐
25. Describe the completion of internal documentation to record service/maintenance provided to tour vehicles.	☐	☐
26. Why is it important to adhere to requirements in relation to the completion and maintenance of vehicle service and repair documentation?	☐	☐

Written questions

Carry out vehicle maintenance or minor repairs – D2.TTO.CL4.02

Student Name: _____

Answer all the following questions and submit to your Trainer.

1. List three ways available to staff to identify the service requirements for vehicles.

2. Identify three examples of situations in which vehicle maintenance may need to be carried out.

3. Describe five activities inherent in a visual inspection of a vehicle for the purpose of providing maintenance.

4. What is the name of the document which should be used when inspecting a vehicle to record the findings of the inspection?

5. Describe three points to note regarding the appearance of a vehicle with the name of a business on a tour vehicle.

6. Describe the process/activities involved in inspecting the panels of a vehicle.

7. What are the four basic inspection procedures for a vehicle chassis as presented in the notes?

8. Identify the three basic diagnostic tools/assessments to be used when inspecting a vehicle chassis.

9. Name six examples of practical maintenance which may be required for a vehicle suspension system.

10. List six checks which need to be made on vehicle headlights when inspecting a vehicle.

11. List four checks which need to be made on vehicle brake lights when inspecting a vehicle.

12. Name six possible adjustments which may be required in relation to the lights on a vehicle.

13. Name four items which need to be inspected when undertaking checks on vision-related items on a vehicle.

14. Identify three examples of action which may need to be taken in response to identified problems with vision-related items on a vehicle.

15. Give four examples of checks which may need to be made when inspecting the entrances and exits of a vehicle.

16. List six examples of basic adjustments for vehicle entrances and exits.

17. Detail four points to note when inspecting the seats of a vehicle.

18. Detail four points to note when inspecting the seatbelts of a vehicle.

19. Give six examples of adjustments which may be necessary for the interior of a vehicle.

20. Describe ten checks which need to be made when inspecting the brakes on a vehicle.

21. Provide six examples of adjustments which may be necessary in response to identified faults with the brakes on a vehicle.

22. Give six examples of checks which may need to be made on the steering system of a vehicle.

23. Provide six examples of adjustments which may be necessary in response to identified faults with the steering on a vehicle.

24. Give six checks which need to be undertaken when inspecting the exhaust system of a vehicle.

25. Describe four activities which may be necessary in response to identified faults with the exhaust system on a vehicle.

26. Describe six checks which need to be undertaken when inspecting towing connections on a vehicle.

27. List four adjustments which may be necessary in response to identified faults with the towing connections on a vehicle.

28. Identify one 'miscellaneous' item that may be inspected on a vehicle and describe six checks which need to be made for that item.

29. For the item identified in the answer to the previous question, list four possible adjustments which may need to be made to effect required service.

30. Identify one 'miscellaneous' item different to the one used in your response to Question 28 that may be inspected on a vehicle and describe six checks which need to be made for that item.

31. For the item identified in the answer to the previous question, list four possible adjustments which may need to be made to effect required service.

32. Describe four checks which may need to be made in relation to load restraints of a vehicle.

33. List four adjustments which may be necessary in response to identified faults with load restraints of a vehicle.

34. List six items which may need to be checked when inspecting the on-board emergency supplies and equipment on a vehicle.

35. Name five Issues which can influence the need to take a vehicle to an external provider for service.

36. Describe six safety requirements which apply in relation to servicing vehicles.

37. What are three basic ways to identify vehicle faults?

38. Describe six SOPs for diagnosing the cause of vehicle faults.

39. Identify six important points to note when undertaking minor repairs to vehicles.

40. Select one of the following and provide the basic steps involved in the process: (1) Replacing globes (2) Replacing a fan belt.

41. Select one of the following and provide the basic steps involved in the process: (1) Replacing spark plugs (2) Replacing glow plugs

42. Give four reasons it is necessary to use correct tools and follow manufacturer's instructions when repairing vehicles.

43. Identify and describe role of one common hand tool used to repair/service a vehicle.

44. Identify and describe role of one common hand tool used to repair/service a vehicle.

45. Select and describe the role of one tyre tool involved in repairing/servicing a vehicle.

46. What is a torque wrench used for?

47. What are multimeters used for?

48. What is the name of the tool commonly used to remove/refit an oil filter?

49. Draw and identify the message of one standard workplace warning sign.

50. Describe six activities inherent in standard practice when completing vehicle inspection checklists.

51. Identify four actions involved in completing internal documentation to record the provision of service and maintenance.

52. Describe three generic sets of requirements which might apply to the completion and maintenance of vehicle repair and service documentation.

Answers to written questions

Carry out vehicle maintenance or minor repairs – D2.TTO.CL4.02

The following are model answers only – Trainers/Assessors must use discretion when determining whether or not an answer provided by a Student is acceptable or not.

1. List three ways available to staff to identify the service requirements for vehicles.

- Researching/reading manufacturer requirements in relation to routine servicing periods and items to be checked and serviced as prescribed in manuals
- Identifying internal vehicle service periods and requirements that may override stated manufacturer's requirements
- Referring to documentation relating to previous servicing
- Talking with vehicle operators
- Identifying host country legislation that applies to the maintenance of commercial vehicles used for passenger transport.

2. Identify three examples of situations in which vehicle maintenance may need to be carried out.

- Day or night
- All weather conditions
- In tight/confined spaces
- In controlled/closed spaces
- In uncontrolled/open environments.

3. Describe five activities inherent in a visual inspection of a vehicle for the purpose of providing maintenance.

- Looking at and under the vehicle and closely checking all aspects of the vehicle
- Lighting the areas being inspected
- Using a checklist
- Inspecting in pairs
- Looking for signs of damage to the vehicle
- Checking for missing parts
- Noting leaking fluids
- Identifying projections from the vehicle
- Taking immediate action to rectify problems
- Checking condition of vehicle instruments.

4. What is the name of the document which should be used when inspecting a vehicle to record the findings of the inspection?

Vehicle inspection checklist

5. Describe three points to note regarding the appearance of a vehicle with the name of a business on a tour vehicle.

- The vehicle nearly always has the name of the tour company on it – so people judge the business by the way the vehicle looks (and the way it is driven)
- Tour group members expect their tour vehicles to be clean and properly maintained when they commence a tour
- Cleanliness is often a major aspect of service and maintenance.

6. Describe the process/activities involved in inspecting the panels of a vehicle.

- Inspect in a systematic fashion
- Use a checklist to record observations
- Identify and note the location and type of any damage observed
- Determine if obvious panel damage is causing/has already caused operational issues for other components of the vehicle
- Work out if the panel damage has the potential to cause operational issues or further damage to other components
- Hammer out damage which is preventing proper use of the vehicle – or using a pull bar or suction device
- Unbolt or cut away panels which are seriously damaged and present a clear threat to the further operation and/or safety of the vehicle.

7. What are the four basic inspection procedures for a vehicle chassis as presented in the notes?

- Wash/clean the chassis/vehicle
- Get the vehicle off the ground – a hoist is preferable
- Remove the wheels and tyres
- Focus on corners, wheel arches and bottom of doors.

8. Identify the three basic diagnostic tools/assessments to be used when inspecting a vehicle chassis.

- Visual assessment – by getting under the vehicle or raising it on a hoist
- Aural assessment – by listening to the vehicle as it travels over uneven ground
- Functional assessment – by gauging the operation/performance of the vehicle on-the-road.

9. Name six examples of practical maintenance which may be required for a vehicle suspension system.

- Adjustment of shock absorbers
- Greasing ball joints, tie rod ends, sway bar links, drag links, centre links, idler arms and Pitman arms
- Inspecting ball joints for play
- Checking wear indicators
- Checking and greasing bearings
- Removing and replacing wheel bearings
- Replacing ball joints and boots
- Checking tyre wear and wheel alignment.

10. List six checks which need to be made on vehicle headlights when inspecting a vehicle.

- Correct colour
- Globes not blown
- High beam – they have good strength and accurate alignment/focus, and illuminate when activated
- Dipped – they have good strength and accurate alignment, and dip and return to high beam when activated: high beam lamps must extinguish when lights are dipped
- Lenses clean and not tarnished or cracked
- Lights are correctly aimed
- No water damage
- Assembly is secure and complete.

11. List four checks which need to be made on vehicle brake lights when inspecting a vehicle.

- Correct colour (red)
- Positioned at correct height
- Activated when brake pedal is depressed
- Clean and clear
- No white light shows through the red cover of the light
- Lenses in good condition.

12. Name six possible adjustments which may be required in relation to the lights on a vehicle.

- Replace blown globes
- Clean lenses and reflectors
- Align/focus headlights, driving lights and fog lights as/if required
- Replace wiring or re-wire lights as required to conform to requirements
- Replace fuses
- Replace wiring
- Replace connectors
- Tighten loose fittings.

13. Name four items which need to be inspected when undertaking checks on vision-related items on a vehicle.

- Glazing/windcreens/windows
- Sun visors
- Windscreen washers and wipers
- Demisters
- Rear vision mirrors – internal and external.

14. Identify three examples of action which may need to be taken in response to identified problems with vision-related items on a vehicle.

- Cleaning
- Tightening or re-affixing connections and fixtures
- Replacement of faulty or defective items
- Professional service by qualified technician to electrical/electronic items.

15. Give four examples of checks which may need to be made when inspecting the entrances and exits of a vehicle.

- The operation of all doors of the vehicle
- The operation of other opening panels/parts of the vehicle
- Aspects of the vehicle allied to entrances and exits
- Emergency exits.

16. List six examples of basic adjustments for vehicle entrances and exits.

- Cleaning
- Removing obstacles
- Greasing and oiling moving parts
- Using dry lube for door strikers
- Using powdered graphite on door locks
- Replacing batteries on remote locking devices/keys
- Replacing seals
- Changing blown globes
- Re-gassing struts
- Providing scheduled maintenance to pneumatic and mechanical devices as prescribed by the manufacturer
- Tightening or re-affixing connections and fixtures
- Replacing faulty or defective items
- Obtaining professional service by qualified technician to mechanical and/or pneumatic items.

17. Detail four points to note when inspecting the seats of a vehicle.

- Seat anchorage points are not cracked or rusted
- There are no missing parts/fasteners in the seat frames or anchorage points
- There are no exposed/sharp edges or protrusions – which may cut/harm people
- Every seat has a seat cushion and backrest – and will provide basic/required level of comfort for passengers
- Seat adjustments (for example, seat height and angle, lumbar support, position of backrest, forward and rear movement/slides) are fully functional – and hold the designated position as required
- Required space around (in front of and in between) seats is provided – in accordance with the type and size of the vehicle as prescribed by relevant transport authority.

18. Detail four points to note when inspecting the seatbelts of a vehicle.

- Seat belt anchorage points are not cracked, damaged, distorted or rusted
- No part of the unit is missing
- The retractor mechanism, locking mechanism, buckle, tongue or adjustment device is working correctly and effectively
- Seatbelt webbing is not absent or damaged/defective in any way.

19. Give six examples of adjustments which may be necessary for the interior of a vehicle.

- Cleaning including removing leaves, dirt and debris from component parts
- Repairing and/or replacing torn/damaged seating, carpet, floor coverings and other internal fabrics/coverings
- Tightening loose fittings
- Replacing deployed or malfunctioning airbags
- Replacing worn, faulty or perished hoses and connections
- Removing items placed over airbag covers
- Replacing seatbelts which have been used when the vehicle was involved in an accident
- Draining hoses and units
- Re-charging air conditioning units with gas – or making arrangements for same
- Re-priming heating systems
- Acquiring, replacing and/or (re-) affixing signs.

20. Describe ten checks which need to be made when inspecting the brakes on a vehicle.

- Ensure brake pedal is properly covered with rubber and no metal showing through
- Verify rubber on pedal has anti-slip surface and is sufficient and in good condition
- Ensure the brake pedal or the hand brake is not physically broken
- Check travel of the brake pedal when applied and ensure it is within manufacturer specifications – for example, less than 20% travel
- Determine if the brake pedal moves to the floor/towards the floor (or if the brake indicator/warning light comes on) when steady pressure is applied for 10 seconds
- Ensure brake warning light illuminates when engine is turned on
- Ensure hand brake warning light illuminates when hand brake is applied – and extinguishes when the hand brake is released
- Make sure the hand brake stays locked in position when applied
- Make sure brake pedal and hand brake return to fully released position when not being used
- Test brakes to ensure they slow/stop vehicle within stated requirements
- Inspect the brake pipes, hoses and connections for signs of damage or severe deterioration, kinks, crimping, heat damage or visible signs of leakage, swelling or bulging
- Inspect the brake pipes, hoses and connections to verify they are securely mounted, and not cracked or broken
- Inspect visible brake components looking for signs they are broken, excessively worn, leaking, contaminated or not securely mounted

- Verify hydraulic lines are of sufficient length to allow for the full range of steering and suspension movement, and are not twisted
- Check any repairs made to hydraulic lines have been made in accordance with applicable standards (for example, they have not been made through heating or welding, and parts used comply with required ratings/standards)
- Check brake fluid and verify it is at required level
- Check visible brake lining material to ensure it is worn to less than manufacturers limits or if the limits are not known 0.8mm above bonded shoe or pad mounting surface and level with the rivet or bolt heads on riveted or bolted lining
- Inspect power/vacuum assistance for brakes to make sure they are operating properly
- Check to ensure compressors, vacuum pumps and/or pulley belts are not cracked, loose or worn
- Look for evidence of leaks from the system (any joint, component or seal).

21. Provide six examples of adjustments which may be necessary in response to identified faults with the brakes on a vehicle.

- Replace rubber on the brake pedal
- Replace broken, damaged, defective, worn or suspect parts – as opposed to repairing them
- Take the vehicle to an auto electrician – to have electrical/warning light problems or frayed electrical wiring replaced/fixed
- Top-up brake fluid
- Replace leaking seals using required kits
- Arrange for, or perform, machining and/or re-conditioning of brake drums and brake disc rotors
- Adjust the hand brake – which may require renewing cables
- Replace brake pads (disc brakes) or shoes (drum brakes)
- Replace the brake fluid (bleed the brakes and re-fill the system)
- Replace wheel cylinder (drum brakes)
- Replace the master cylinder and servo unit
- Replace brake hoses
- Adjust disc brakes.

22. Give six examples of checks which may need to be made on the steering system of a vehicle.

- With the engine running, check the operation of the steering by moving the steering wheel – to ensure:
 - There is no more than 50 mm rotational free play in the wheel
 - The steering wheel is free from structural damage
 - Any accessories fitted to steering wheels are not loose
 - The steering wheel is securely attached to the steering column
 - Where a SRS airbag is fitted, there is any evidence airbags or other SRS systems are operative: check the indicator light, where fitted.
- Visually inspect steering components under the bonnet and vehicle – to ensure:
 - No steering component is missing, cracked or broken or is worn beyond manufacturer's limits
 - No steering component can be seen to have been repaired or modified by bending, heating or welding
 - No nut, bolt or locking device is missing or insecure
 - Tie rod and drag link ends are secured in both the rod and taper with fasteners suitably locked (e.g. split pins, lock-wire, tabs or self-locking nuts)
 - There is no movement on the spline between Pitman arm and the steering box or between any thread or tapered joint
 - No free play due to wear in any steering component which exceeds manufacturer's specification
 - Power steering has no leaks, no defects and is not inoperative
 - Any manual or power steering componentry is securely mounted and free from excessive side or end play, roughness, or binding
 - Power steering belts are not loose, broken, frayed, missing, or cracked through to reinforcing plies.
- Check the operation of the anti-theft/steering lock – to ensure:
 - The ignition key cannot be removed in any position except the "anti-theft" (lock) position
 - When engaged, the anti-theft lock prevents at least one of the following actions:
 - Steering the vehicle
 - Engaging the forward drive gears
 - Release of the brakes.

23. Provide six examples of adjustments which may be necessary in response to identified faults with the steering on a vehicle.

- Arrange for professional repairs to be made where serious issues are identified and/or where work involving airbags needs to be done
- Replace worn/loose steering wheel covers
- Replace damaged steering wheels
- Tighten nuts, bolts, connections, fasteners and steering wheel accessories
- Replace defective/leaking seals
- Replace missing, cracked, broken or excessively worn components
- Top-up power steering fluid
- Secure tie rod and drag link ends where required
- Replace defective/damaged hoses, pipes and drive belts
- Adjust and replace power steering drive belt
- Adjust toe alignment on wheels
- Lubricate steering swivel joints
- Replace track rods and (track rod end) ball joints
- Set the tracking.

24. Give six checks which need to be undertaken when inspecting the exhaust system of a vehicle.

- There are no visible cracks
- The system is securely mounted – across its entire length
- An effective silencer if fitted and operational with all exhaust gases passing through it – so vehicle noise is not excessive in accordance with local regulations/mandated noise levels
- Outlet of exhaust pipe is to rear of passenger side doors and opening windows
- Exhaust discharges:
 - Rear of vehicle
 - To the correct side of the vehicle (left or right) depending on which side of the road the vehicle is driven on
- No leaks, holes or corrosion in the system
- The exhaust pipe exit extends *beyond* the vehicle body but not beyond the overall plan of the vehicle
- The system does not interfere with or obstruct any part of the steering, suspension, brake or fuel system
- The *entire* system at its lowest points clears the road surface by the required distance – for example, 100 mm
- The exhaust does not let out sparks, flames, excessive gases, oil or fuel residue

- No visual emissions from the exhaust for longer than 10 seconds after the engine has been started
- Where fitted, emission control equipment is all present and fully operational
- LPG fuel lines are separated from exhaust components by the required distance (say 150 mm) and/or protected by a shield and separated by a lesser distance, say 50 mm.

25. Describe four activities which may be necessary in response to identified faults with the exhaust system on a vehicle.

- Replacing seals
- Replacing mountings
- Replacing damaged, sub-standard and defective parts – rather than repairing them, or patching leaks/holes
- Attaching missing parts/components
- Re-routing the system – to comply with legally imposed requirements
- Tuning the vehicle.

26. Describe six checks which need to be undertaken when inspecting towing connections on a vehicle.

- There are no cracks or other visible physical damage in any part of the towing attachment
- The towing attachment is secure/not loose
- Towing bar is fastened securely to the vehicle
- No visible corrosion on bolts/any part of the towing connection
- Towing connection is suitable for the weight to be towed – and the vehicle displays (where and as required) an appropriate label to this effect
- No temporary repairs or welds (especially where drawbar is marked 'Do Not Weld' by the manufacturer) have been made to the connection
- Electrical wiring, couplings and connectors are securely fastened and in good condition
- An effective secondary locking system for the trailer
- Tow-ball assembly is securely attached
- Safety chains and/or cables are in good condition – and:
 - Can be affixed so they cannot be accidentally detached
 - Are securely attached to the drawbar
- Safety chains are rated for the weight to be drawn – and in the required number
- All required parts of detachable tow units are in position/present in the assembly
- Jockey wheel is in position, firmly attached and fully functional.

27. List four adjustments which may be necessary in response to identified faults with the towing connections on a vehicle.

- Replacing defective parts – rather than repairing them
- Removing un-rated chains and replacing them with chains which are stamped and correctly rated – for the intended towing weight
- Cleaning and greasing the jockey wheel.

28. Identify one ‘miscellaneous’ item that may be inspected on a vehicle and describe six checks which need to be made for that item.

No model answer.

29. For the item identified in the answer to the previous question, list four possible adjustments which may need to be made to effect required service.

No model answer.

30. Identify one ‘miscellaneous’ item different to the one used in your response to Question 28 that may be inspected on a vehicle and describe six checks which need to be made for that item.

No model answer.

31. For the item identified in the answer to the previous question, list four possible adjustments which may need to be made to effect required service.

No model answer.

32. Describe four checks which may need to be made in relation to load restraints of a vehicle.

- Load anchorages – inspecting and testing to:
 - Ensure they are securely mounted and have not worked loose
 - Make sure there is no visible rust/corrosion to the anchorages or surfaces to which they are attached
- Retention devices, baggage and cargo restraints – inspecting and testing to:
 - Ensure they remain securely fitted and have not worked loose
 - Verify operation as intended
 - Confirm effectiveness and suitability for loads/items to be secured/restrained
 - Make sure there is no corrosion to steel components
 - Check condition of straps, webbing and nets – no rips, missing parts, damage, deterioration
 - Count/examine number of cords/straps to ensure they are all present
 - Confirm operation of ratchets and rollers
 - Make sure D-rings and other attachments/connectors are in place, undamaged and fully-functional

- Cab guards – these are the devices designed to protect people in the cab of a vehicle from intrusion of items coming in through the rear window during an accident or a quick stop/hard braking. Checks should verify:
 - Mounting points are secure and solid
 - Unit is rigid and fully covering the window being protected
 - No corrosion/rust near mounting/anchor points
 - No damage to the unit which compromises its integrity
 - No loose parts
 - Mesh/wire has not broken/become unattached and become a hazard for people to cut themselves on
 - Required vision from the cab is not obstructed
- Headboards, sideboards and tailboards – inspecting and testing to ensure :
 - Operation as intended – opening and closing/moving up and down
 - Boards are securely attached to the vehicle itself
 - Boards can be securely locked in position
 - No fasteners are missing or loose
 - No evidence of rust or corrosion – at mounting points and on boards themselves
 - No evidence of cracks or fractures
 - No proof of significant damage or distortion – which impacts function and/or presents a hazard

Curtain systems – these are the flexible curtains commonly used on the side of vehicles/trucks to contain loads. Checks need to ensure:

- They are not ripped/torn or have not otherwise deteriorated or been damaged
- They are securely attached to the frame which supports them – and the frame is securely attached to the vehicle
- Previous repairs made to the fabric/material remain satisfactory
- Correct rating system/curtain is being used (where applicable) – and relevant certificate is available
- Curtain tensioning components are fully functional, secure and in good condition.

33. List four adjustments which may be necessary in response to identified faults with load restraints of a vehicle.

- Cleaning
- Welding broken/damaged parts
- Replacing straps, webbing and nets
- Replacing anchor points, bolts and/or other connectors
- Repairing and/or replacing boards and curtains
- Replacing defective parts – where repairs cannot be successfully undertaken
- Greasing moveable parts.
- Fitting items where revised use of vehicles indicates need to do so.

34. List six items which may need to be checked when inspecting the on-board emergency supplies and equipment on a vehicle.

- Jacks
- Tool kit
- Spare parts
- Spare tyre/s
- Tyre chains
- Tyre studs
- Recovery equipment
- Fire extinguishers
- First aid kits
- EPIRBs.

35. Name five Issues which can influence the need to take a vehicle to an external provider for service.

- Extent of work which needs to be done
- Timeframes which apply
- Safety issues
- Complexity of the job
- Tools and equipment available
- Expertise
- Terms of warranties and guarantees'.

36. Describe six safety requirements which apply in relation to servicing vehicles.

- Taking responsibility for personal safety
- Being alert to the possibility of risk
- Warning others of hazards – so they avoid them
- Learning workplace safety policies and procedures
- Understanding a hazard can exist today that did not exist yesterday
- Realising a remedy for one workplace risk can introduce another hazard
- Reporting any workplace risk/hazard – as soon as it is identified
- Keeping the work area neat and tidy
- Wearing clothes that are fit for the job
- Reporting 'near misses' as well as actual accidents/injury
- Looking out for the safety of others – other staff and members of the public
- Immediately ceasing any work as soon as it becomes dangerous
- Participating in workplace safety discussions, inspections, feedback and management – so workplace safety is consultative
- Adhering to SOPs, SWPs and manufacturer's instructions
- Wearing/using designated PPE as directed
- Using all appliances, equipment, tools as directed
- Intervening when a dangerous workplace situation is identified
- Taking care when working
- Picking up items from the floor when they have been dropped
- Cleaning up spills immediately
- Disposing of glass (including broken glass) into designated containers
- Undertaking a first aid course
- Being trained in workplace health and safety
- Not working when unfit to do so – as a result of tiredness, injury, other causes
- Seeking to establish a safe place/working environment
- Applying safe manual handling protocols
- Ensuring there are MSDS for all chemicals used in the workplace
- Handling chemicals according to established safe chemical handling, storing and usage protocols
- Abiding by the requirements of health, safety and welfare legislation
- Including health and safety training as part of the Induction and Orientation
- Following manufacturer's instructions
- Ensuring a fire extinguisher is close-by – just in case it is needed.

37. What are three basic ways to identify vehicle faults?

- Talking with those who have operated/driven or been in a vehicle
- Test driving the vehicle
- Running stationary vehicle to induce fault
- Reading reports which have been prepared on the vehicle.

38. Describe six SOPs for diagnosing the cause of vehicle faults.

- Analysing available information ('evidence') as obtained from the fault identification process
- Using diagnostic flowcharts
- Using on-board diagnostics (OBD)
- Referring to service and operator manuals
- Seeking additional advice from the manufacturer
- Over the phone assistance – from the manufacturer or the agent/business that sold the vehicle
- Referring to personal experience
- Loading vehicle onto jacks, stands or a hydraulic lift – in order to:
 - Get a better look at the vehicle and be able to see 'up and under' it and the engine bay. Certainly, raised vehicles always make it easier to identify the location of leaks and drips.
 - Free up certain suspension components
 - Make it easier to manipulate other parts of the vehicle
- Using applicable sensory appraisal to help determine cause
- Monitoring on-board instrument panel
- Monitoring externally attached equipment.

39. Identify six important points to note when undertaking minor repairs to vehicles.

- Performing repairs to vehicles which are still under warranty can void the warranty – so there is a need to check this before proceeding any further
- The actual work required may differ between makes and models of vehicles – the procedures listed in the notes are indicative only
- Always be prepared to read Owner's manual or Workshop manual
- Obtain the right parts before starting the job
- Obtain and use the right tool/s for the job
- Work steadily – never rush a job
- Check the surrounding areas/sections for debris and/or damage at the same time – and respond as indicated
- Never be afraid of asking for help, advice or direction from a more qualified or experienced person.

- 40. Select one of the following and provide the basic steps involved in the process: (1) Replacing globes (2) Replacing a fan belt.**

No model answer.

- 41. Select one of the following and provide the basic steps involved in the process: (1) Replacing spark plugs (2) Replacing glow plugs**

No model answer.

- 42. Give four reasons it is necessary to use correct tools and follow manufacturer's instructions when repairing vehicles.**

- Save time
- Save damage to the item being fitted and/or vehicle
- Reduce potential for personal injury
- Retain parts warranties and guarantees
- Optimise service life of part being fitted
- Ensure part integrates fully/properly with other items.

- 43. Identify and describe role of one common hand tool used to repair/service a vehicle.**

No model answer.

- 44. Identify and describe role of one common hand tool used to repair/service a vehicle.**

No model answer.

- 45. Select and describe the role of one tyre tool involved in repairing/servicing a vehicle.**

No model answer

- 46. What is a torque wrench used for?**

Used to tighten a fastener to a specific amount of force

- 47. What are multimeters used for?**

Used to measure voltage, resistance, and amperage

- 48. What is the name of the tool commonly used to remove/refit an oil filter?**

Oil filter wrench

- 49. Draw and identify the message of one standard workplace warning sign.**

No model answer

50. Describe six activities inherent in standard practice when completing vehicle inspection checklists.

- Obtain the necessary checklist
- Carry and use the checklist while inspecting the vehicle
- Undertake the inspection at the required times
- Inspect the vehicle in the sequence of the checks listed on the checklist
- Undertake all the listed inspections – never skip any of them
- Inspect the vehicle in the ways required by the checklist
- Tick off the items (✓) as they are inspected – to show they have been checked and are acceptable/OK
- Place a cross (X) in the tick boxes where a problem is identified – and add some written comment
- Place unsafe/un-roadworthy vehicles out-of-service – so they cannot be used
- Write down the odometer
- Date the completed checklist – to give a point of reference
- Sign the form (there may also be a need to print name)
- Process the completed form.

51. Identify four actions involved in completing internal documentation to record the provision of service and maintenance.

- Using the approved template/form
- Identifying the vehicle which has been serviced/repaired
- Completing the form immediately after the work has been completed – or as the work progresses
- Describing service and repairs provided
- Noting matters that will/may require attention at the next service
- Recording the action taken for cost and stock control purposes
- Updating next scheduled service for the vehicle
- Attaching the inspection checklist (or copy) which initiated/triggered the work to the completed 'Service Report'
- Returning the vehicle to its normal location – to return it to service
- Notifying personnel who may be waiting on the vehicle – that it is now ready.

52. Describe three generic sets of requirements which might apply to the completion and maintenance of vehicle repair and service documentation.

- A range of host enterprise requirements - which can relate to:
 - Monitoring costs
 - Tracking vehicle use
 - Identifying vehicle service dates
 - Identifying vehicle dates
 - Calculating economy
- Need to comply with legislated requirements – especially:
 - Vehicle registration in the correct ('commercial') category
 - Mandatory insurance
 - Roadworthiness
- Requirements imposed on the business by clients such as tour operators, venues and site operators who contract the use of vehicles – as these apply to:
 - Cleaning of vehicles ready for use
 - Maintaining service of vehicles for operational reliability and safety
 - Ensuring availability of vehicle at times they are required/booked/contracted
 - Recording use by external parties
 - Capturing usage facts for administration and charging purposes
- Requirements imposed on the business by third parties such as insurers and auditors – such as:
 - Advising of incidents and accidents, as/if necessary
 - Notifying of repairs and/or modifications
 - Maintaining insurance coverage
 - Ensuring vehicles are maintained in a condition (operational, appearance, comfort, safety, facilities) that complies with the standard demanded by the code/system being audited
- Taxation agency for the country – in terms of:
 - Maintaining records to show usage information of the vehicle/distance travelled and other data as may be required to demonstrate legitimate business-related use – for example:
 - Times and dates
 - Destinations
 - Purpose/reason for travel
 - Retain evidence/receipts of work performed on vehicle and fuel for it.

Observation checklist

Student name	
Assessor name	
Location/venue	
Unit of competency	Carry out vehicle maintenance or minor repairs D2.TTO.CL4.02
Dates of observation	
Instructions	- Over a period of time observe the student completing each of the following tasks: - Provide scheduled service to vehicles - Diagnose minor vehicle faults - Undertake minor repairs to vehicles - Complete documentation - Enter the date on which the tasks were undertaken - Place a tick in the box to show they completed each aspect of the task to the standard expected in the enterprise - Complete the feedback sections of the form, if required.

Did the candidate	Yes	No
Element 1: Provide scheduled service to vehicles		
Identify necessary minor servicing requirements for specific vehicles	☐	☐
Determine situations in which maintenance may need to be carried out	☐	☐
Undertake visual inspections of the vehicle	☐	☐
Check and adjust vehicle structure	☐	☐
Check and adjust lighting	☐	☐
Check and adjust vision-related items	☐	☐
Check and adjust entrances and exits	☐	☐
Check and adjust vehicle interior	☐	☐
Check and adjust brakes	☐	☐
Check and adjust steering	☐	☐

Did the candidate	Yes	No
Check and adjust exhaust	☐	☐
Check and adjust towing connections	☐	☐
Check and adjust miscellaneous items	☐	☐
Check and adjust load restraints	☐	☐
Check and replenish on-board emergency equipment and supplies, where necessary	☐	☐
Make arrangements for external service provision where requirements cannot be accommodated internally	☐	☐
Comply with mandated safety requirements while undertaking service	☐	☐
Element 2: Diagnose minor vehicle faults		
Identify minor faults	☐	☐
Determine cause of faults	☐	☐
Element 3: Undertake minor repairs to vehicles		
Remove, repair and/or refit vehicle components	☐	☐
Use correct tools and follow manufacturer's instructions	☐	☐
Make arrangements for external repair provision where requirements cannot be accommodated internally	☐	☐
Comply with mandated safety requirements while undertaking repairs	☐	☐
Element 4: Complete documentation		
Ensure vehicle inspection checklists are completed	☐	☐
Complete internal documents to record provision of service and repairs	☐	☐
Adhere to requirements in relation to the completion and maintenance of vehicle service and repair documentation	☐	☐
Did the student's overall performance meet the standard?	☐	☐

Feedback to student and trainer/assessor**Strengths:****Improvements needed:****General comments:****Candidate signature****Date****Assessor signature****Date**

Third Party Statement

Student name:			
Name of third party:		Contact no:	
Relationship to student:	☐ Employer ☐ Supervisor ☐ Colleague ☐ Other <i>Please specify: _____</i> <i>Please do not complete the form if you are a relative, close friend or have a conflict of interest]</i>		
Unit of competency:	Carry out vehicle maintenance or minor repairs D2.TTO.CL4.02		
The student is being assessed against industry competency standards and we are seeking your support in the judgement of their competence. Please answer these questions as a record of their performance while working with you. Thank you for your time.			
Do you believe the trainee has demonstrated the following skills? <i>(tick the correct response]</i>	Yes	No	Not sure
Identifies minor vehicle servicing requirements and situations	☐	☐	☐
Performs standard vehicle inspections	☐	☐	☐
Checks and adjusts basic operating components identified as a result of standard vehicle inspections and scheduled service requirements	☐	☐	☐
Identifies and diagnoses basic minor vehicle faults	☐	☐	☐
Rectifies and/or arranges for repair of minor vehicle faults	☐	☐	☐
Complies with manufacturer's instructions and mandated safety requirements when undertaking repairs	☐	☐	☐
Completes inspection checklists as part of vehicle inspection/s, service and repair/s	☐	☐	☐

Comments/feedback from Third Party to Trainer/Assessor:

Third party signature:

Date:

Send to:

Competency recording sheet

Name of Student		
Name of Assessor/s		
Unit of Competency	Carry out vehicle maintenance or minor repairs	D2.TTO.CL4.02
Date assessment commenced		
Date assessment finalised		
Assessment decision	Pass Competent / Not Yet Competent (Circle one)	
Follow up action required (Insert additional work and assessment required to achieve competency)		
Comments/observations by assessor/s		

Place a tick (✓) in the column to reflect evidence obtained to determine Competency of the student for each Performance Criteria.

Element & Performance Criteria	Observation of skills	3rd Party Statement	Oral Questions	Written Questions	Work Projects	Other
Element 1: Provide scheduled service to vehicles						
Identify necessary minor servicing requirements for specific vehicles						
Determine situations in which maintenance may need to be carried out						
Undertake visual inspections of the vehicle						
Check and adjust vehicle structure						
Check and adjust lighting						
Check and adjust vision-related items						
Check and adjust entrances and exits						
Check and adjust vehicle interior						
Check and adjust brakes						
Check and adjust steering						
Check and adjust exhaust						
Check and adjust towing connections						
Check and adjust miscellaneous items						
Check and adjust load restraints						

Place a tick (✓) in the column to reflect evidence obtained to determine Competency of the student for each Performance Criteria.

Element & Performance Criteria	Observation of skills	3rd Party Statement	Oral Questions	Written Questions	Work Projects	Other
Check and replenish on-board emergency equipment and supplies, where necessary						
Make arrangements for external service provision where requirements cannot be accommodated internally						
Comply with mandated safety requirements while undertaking service						
Element 2: Diagnose minor vehicle faults						
Identify minor faults						
Determine cause of faults						
Element 3: Undertake minor repairs to vehicles						
Remove, repair and/or refit vehicle components						
Use correct tools and follow manufacturer's instructions						
Make arrangements for external repair provision where requirements cannot be accommodated internally						
Comply with mandated safety requirements while undertaking repairs						
Element 4: Complete documentation						

Place a tick (✓) in the column to reflect evidence obtained to determine Competency of the student for each Performance Criteria.

Element & Performance Criteria		Observation of skills	3rd Party Statement	Oral Questions	Written Questions	Work Projects	Other
Ensure vehicle inspection checklists are completed							
Complete internal documents to record provision of service and repairs							
Adhere to requirements in relation to the completion and maintenance of vehicle service and repair documentation							
Candidate signature:				Date:			
Assessor signature:				Date:			

